UA Innovate 2026: Using Data to Evaluate Network Costs and Identify Opportunities for Improvement/Optimization

Southern Company

February 28, 2026

Executive Summary

This report summarizes the current network lifecycle posture based on Southern Company infrastructure datasets and a consolidated device-level analytics pipeline. Code at GitHub.

Key Snapshot (Current `data/device_dataset.csv`, mapped locations only)

- Total devices analyzed (mapped): **18,845**
- Unique mapped sites: **1,852**
- States represented: **19**
- Overdue devices (`days_to_eol < 0`): **1,173**
- Devices within 365-day horizon: **758**
- Devices with unknown EoL: **12,444**
- Estimated near-term replacement exposure (overdue + within horizon): **$6.85M**

Count Reconciliation

- Total rows in `data/device_dataset.csv`: **19,235**
- Rows used for map/site-level findings: **18,845**
- Difference (**390**) comes from rows without complete location mapping (`loc_state`, `loc_site_code`, `loc_latitude`, or `loc_longitude` missing), which are excluded from location-based summaries and map visuals.

The analysis indicates concentrated lifecycle backlog at a small set of sites and material replacement exposure concentrated in specific operating regions and model families.

1 Business Objective

Enable leadership to:

- Quantify near-term lifecycle replacement risk and cost exposure.
- Identify locations and model families with outsized backlog.
- Prioritize remediation actions before proceeding to ML-based forecasting and optimization.

2 Data Inputs

Primary Workbook

- `data/UAInnovateDataset-SoCo.xlsx`

Generated Working Dataset

- `data/device_dataset.csv`

Tabs Integrated

- Inventory systems: `CatCtr`, `PrimeAP`, `PrimeWLC`, `NA`
- Site/location metadata: `SOLID`, `SOLID-Loc`
- Lifecycle/cost metadata: `ModelData`, `Pricing`

3 Data Preparation and Filtering Logic

Implemented in:

- `src/utils/data/data_loader.py`

  Core processing steps:

1. Load all workbook tabs (`wrangle`).
2. Apply business filters (`apply_filter`):
   - Remove decommissioned sites from `SOLID` and `SOLID-Loc`.
   - In `CatCtr`, remove unreachable/degraded entries (`Unreachable`, `Ping Reachable`).
   - Remove unsupported rows with missing `hostname` in `CatCtr`.
   - Deduplicate across inventory sources using name matching precedence.
   - Remove inactive NA devices (`Device Status = Inactive`).
   - Exclude NA device types out of scope (`Wireless Controller`, `Firewall`, `Virtual Firewall`, `WirelessLC`).

3. Build unified device dataset (`get_device_dataset`):
   - Normalize identities and map each device to location attributes (`state/site code` + `SOLID`/`SOLID-Loc`).
   - Join lifecycle and replacement metadata from `ModelData`.
   - Join replacement pricing signals from `Pricing`.
4. Clean analytical dataset (`clean_device_dataset`) by dropping low-value/noisy columns.

4 Findings

4.1 National and Site-Level Visibility (Dashboard Map)

The interactive map view supports state and site-level lifecycle triage.

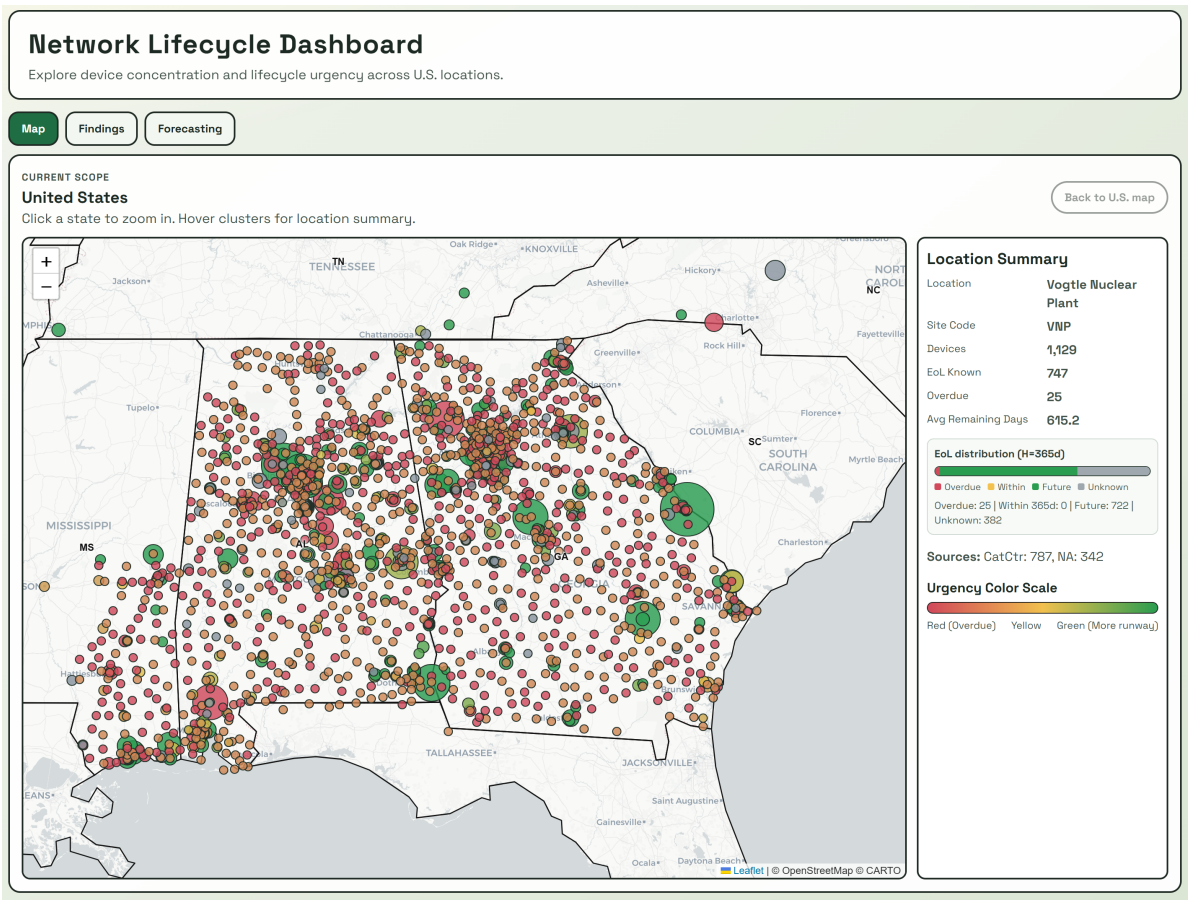

Figure 1: Dashboard map view (national overview with site-level clusters).

4.2 Highest Backlog Sites (Top 10 Overdue)

Overdue lifecycle burden is concentrated in a limited number of sites, supporting targeted intervention over broad, undifferentiated action.

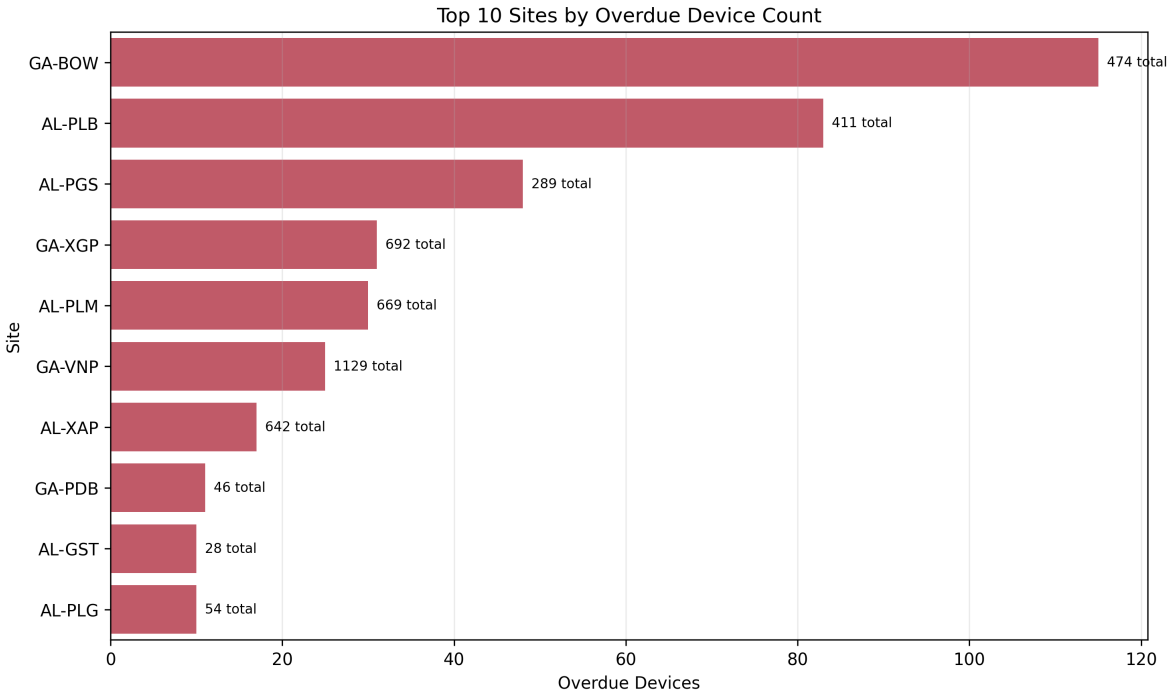

Figure 2: Top overdue sites by device count and near-term cost exposure.

Near-Term Cost Calculation (Used in Top-10 Ranking)

Per-device replacement estimate:

- If `pricing_total_estimate` is available, use it.
- Else use `modeldata_material_cost`.
- Else use `0`.

Near-term scope flag:

- Include device if `days_to_eol < 0` (overdue), or
- Include device if `0 <= days_to_eol <= 365` (within horizon), else exclude.

Near-term contribution per device:

$$\texttt{near_term_cost}_i = 1\{\texttt{in_scope}_i\} \cdot \texttt{replacement_cost}_i$$

Portfolio/site near-term exposure:

$$\texttt{NearTermExposure} = \sum_i \texttt{near_term_cost}_i$$

Note:

- `modeldata_material_cost` is material-side (device + DNA + staging + tax/overhead).
- Labor is tracked separately in `modeldata_labor_cost` and is not currently added into near-term exposure.

Top 10 Overdue Sites (Current Snapshot)

Rank	Site	Overdue Devices	Total Devices	Near-Term Cost
1	GA-BOW (Plant Bowen)	115	474	$263,498.77
2	AL-PLB (APC - Plant Barry)	83	411	$158,117.88
3	AL-PGS (APC - Plant Gaston)	48	289	$116,276.01
4	GA-XGP (GPC - Corp HQ)	31	692	$172,226.39
5	AL-PLM (APC-Miller Steam Plant)	30	669	$93,801.34
6	GA-VNP (Vogtle Nuclear Plant)	25	1,129	$64,309.02
7	AL-XAP (APC - Corp Hdqtrs)	17	642	$46,016.96
8	GA-PDB (Plant Dahlberg)	11	46	$39,840.49
9	AL-GST (GSC#10-Telecom Shack)	10	28	$33,670.40
10	AL-PLG (Plant Gorgas)	10	54	$21,698.65

Table 1: Top 10 overdue sites and near-term replacement exposure.

4.3 Lifecycle vs Support Coverage, Security Risk, and Cost

Site-level risk/cost patterns show clusters where low lifecycle runway aligns with higher security-risk proxy and larger near-term cost pressure.

Security risk proxy definition (device-level):

- `days_to_eol = recommended_eol_date - today`
- Base risk by lifecycle status:
  - `80` if overdue (`days_to_eol < 0`)
  - `50` if within horizon (`0..365`)
  - `20` if future (`>365`)
  - `45` if EoL unknown
- Add `+10` if both software and firmware telemetry are missing
- Final clamp: `risk_i = clamp(base_i + 10*missing_version_i, 0, 100)`

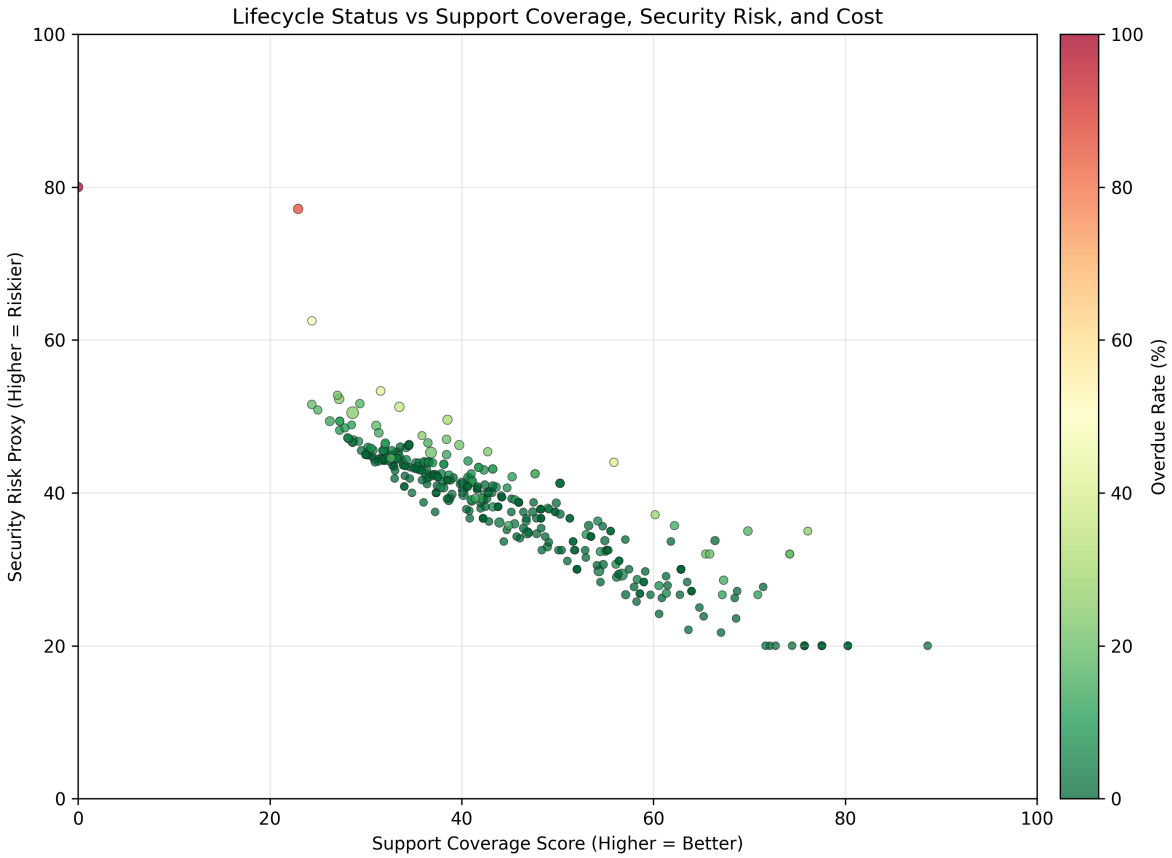

Figure 3: Risk proxy vs near-term cost exposure (site-level).

4.4 Model Hotspots

A subset of models drives disproportionate overdue counts and near-term replacement cost exposure.

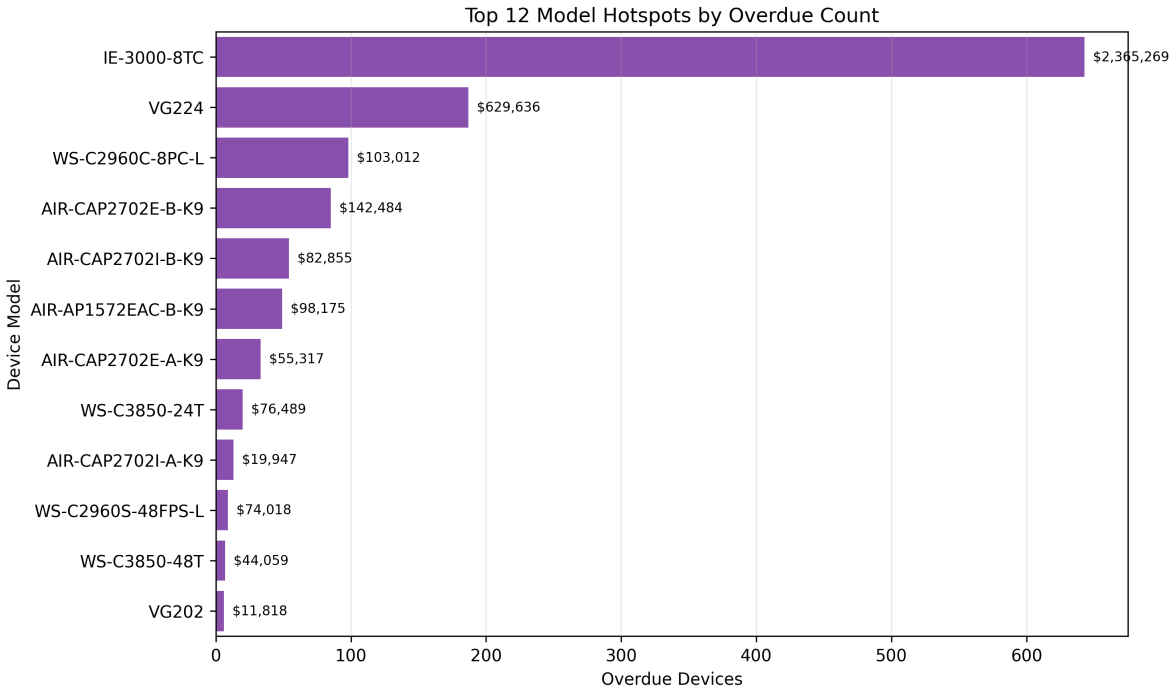

Figure 4: Model families with highest lifecycle backlog and cost exposure.

4.5 Portfolio Lifecycle Distribution

Unknown EoL coverage remains high and is itself a planning risk because it weakens confidence in replacement forecasting.

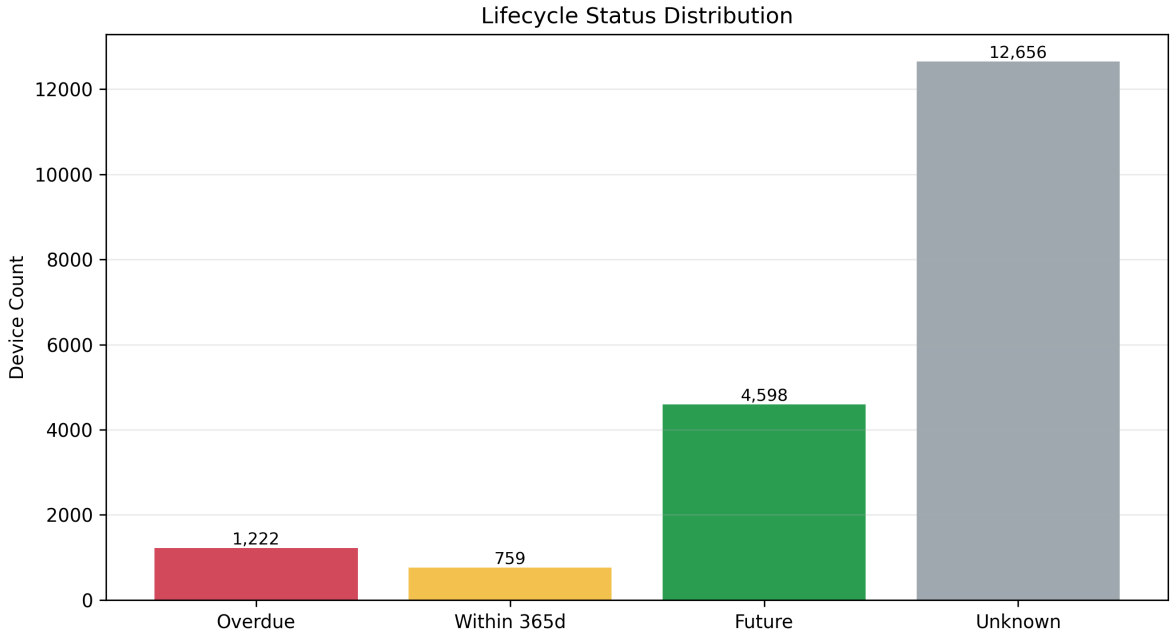

Figure 5: Portfolio lifecycle bucket distribution (including unknown EoL).

5 Risk and Limitations

- EoL coverage is incomplete (large unknown segment), which limits confidence in deterministic replacement scheduling.
- Security risk is a proxy metric derived from lifecycle status and telemetry completeness; it is not a direct vulnerability feed.
- Cost signals are estimated from available pricing/model mappings and should be treated as planning-grade, not final procurement quotes.

6 Business Recommendations (Near-Term)

1. Launch a targeted remediation plan for the highest-backlog sites first (top-10 focus).
2. Establish a data quality sprint to reduce unknown EoL share and improve forecast reliability.
3. Prioritize model families with both high overdue counts and high near-term cost exposure.
4. Use state/site call-group allocation to stage operational execution capacity.

7 Machine Learning Model for 12-Month Expiry Prediction

We implemented an initial machine-learning pilot in `notebooks/02_model.ipynb` to classify devices into:

- `-1`: already passed EoL
- `1`: expected to expire within 365 days
- `0`: EoL more than 365 days away

Modeling Workflow

1. Start from cleaned `data/device_dataset.csv` and keep rows with known EoL labels for supervised training.
2. Build the 3-class target from `days_to_eol`.
3. Train a Random Forest classifier with train-only preprocessing and grouped validation logic to reduce leakage.
4. Evaluate using confusion matrix and ROC/AUC, with additional emphasis on class-wise recall.

Current Result and Observations

- Overall AUC is strong (about **0.875**), but performance is not balanced across classes.
- The model performs best on the majority class (`0: EoL > 365`).
- In some runs, the model produced zero predictions for the passed-EoL class, which is operationally unacceptable.

Next Improvement Direction

- Re-balance training and scoring to prioritize recall for passed-EoL first, then within-12-month devices, and only then the long-runway class.
- Add feature engineering for older device cohorts and calibration/threshold tuning focused on minority critical classes.

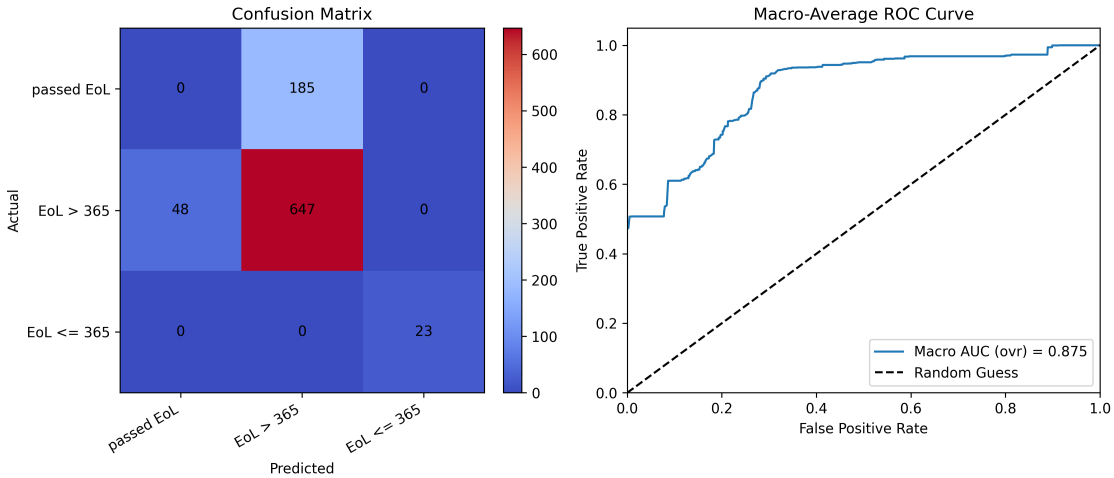

Figure 6: Model performance summary (confusion matrix and macro-average ROC/AUC).

A Appendix Visuals

Additional technical visuals available in `reports/`:

- `appendix_days_to_eol_distribution.png`
- `appendix_overdue_age_distribution.png`
- `appendix_state_exposure.png`
- `appendix_source_lifecycle_mix.png`
- `appendix_category_lifecycle_heatmap.png`
- `appendix_owner_risk_boxplot.png`
- `appendix_call_group_backlog.png`
- `appendix_unknown_eol_by_state.png`
- `appendix_cost_composition.png`
- `appendix_site_priority_matrix.png`